Classifying Shooting Incident Fatality in New York City

Using Machine Learning for Law Enforcement Risk Assessment

MSC DATA SCIENCE · UNIVERSITY OF HERTFORDSHIRE

7PAM2002-0206-2025 DATA SCIENCE PROJECT

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Original Dataset derived : [NYPD_Shooting_Incident_Data__Historic.csv]

GitHub: **https://github.com/indhu0204/NYPD-Shooting-Fatality-Classification**

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Problem Statement & Research Context

> "Can machine learning models accurately predict whether an NYPD shooting incident will result in fatality, and which spatial, temporal, and demographic factors are most strongly associated with fatal outcomes?"

The Challenge

- 27,185 shooting incidents in NYC (2006–2022)
- 19.3% result in fatalities — 5,244 deaths
- Limited resources: trauma teams, investigators, patrol units

Why This Matters

- Enable real-time risk assessment for resource deployment
- Identify geographic hot spots for preventive policing
- Understand environmental vs. demographic factors
- Support evidence-based policy and intervention strategies

Literature Foundation

- Spatial crime concentration (Weisburd, 2015)
- Predictive policing effectiveness (Mohler et al., 2015)
- ML for crime prediction (Wang et al., 2012)
- Class imbalance handling (He and Garcia, 2009)

Dataset & Methodology

Data Source

NYC Open Data Portal – NYPD Shooting Incident Data (Historic).

Accessed data on 23rd january, before it was updated on february 26th.

Dataset: NYPD_Shooting_Incident_Data__Historic.csv

Training: 2006–2022 (27,185 incidents). **Holdout:** 2023–2024 (2,432 incidents) reserved for validation. Final clean records: 27,185 (99.5% retention).

Preprocessing

- Dropped perpetrator features (34% missing – many shooters unidentified)
- Dropped location context fields (94% missing)
- Handled string nulls: 'null', 'UNKNOWN', '(null)'
- Converted coordinates from string to float

Feature Engineering

- Extracted: Hour (0–23), Month (1–12), DayOfWeek (0–6)
- Derived: IsWeekend, Is NightTime (binary flags)
- Encoded: BORO, VIC_AGE, VIC_SEX, VIC_RACE (Label Encoding)

Feature Sets Tested

Full

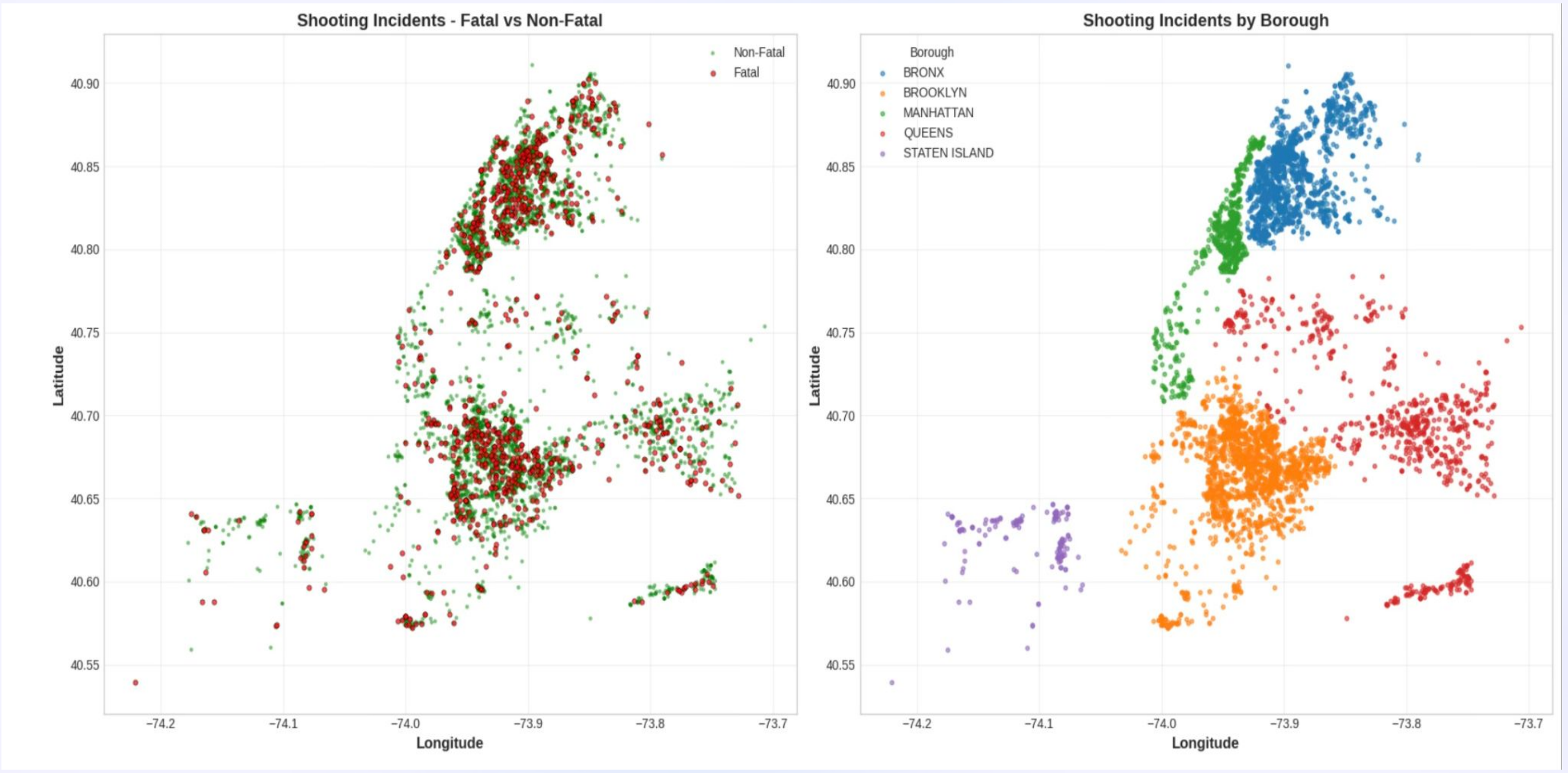
13 features – all geographic, temporal, demographic

Simple_02

5 features – Lat, Lon, Hour, Month, PRECINCT

Simple_03

4 features – Lat, Lon, Hour, Month



Exploratory Data Analysis: Key Findings

Geographic Patterns — Fatality Rate by Borough

Borough	Incidents	Fatality Rate
Staten Island	773	21.0% ▲
Queens	4,070	19.8%
Bronx	7,902	19.4%
Brooklyn	10,882	19.4% (40% of total)
Manhattan	3,558	17.7% ▼

Geographic Clustering Visualization

- Fatal incidents (red dots) concentrate in central Brooklyn/Bronx
- Non-fatal (green) spread more widely across all boroughs
- Clear geographic clustering visible – supports spatial hot spot theory
- Brooklyn (blue) dominates with densest concentration (40% of incidents)
- Staten Island (purple, bottom-left) isolated but high fatality rate

TAKEAWAY: Location matters – fatal cases cluster geographically

Temporal Patterns

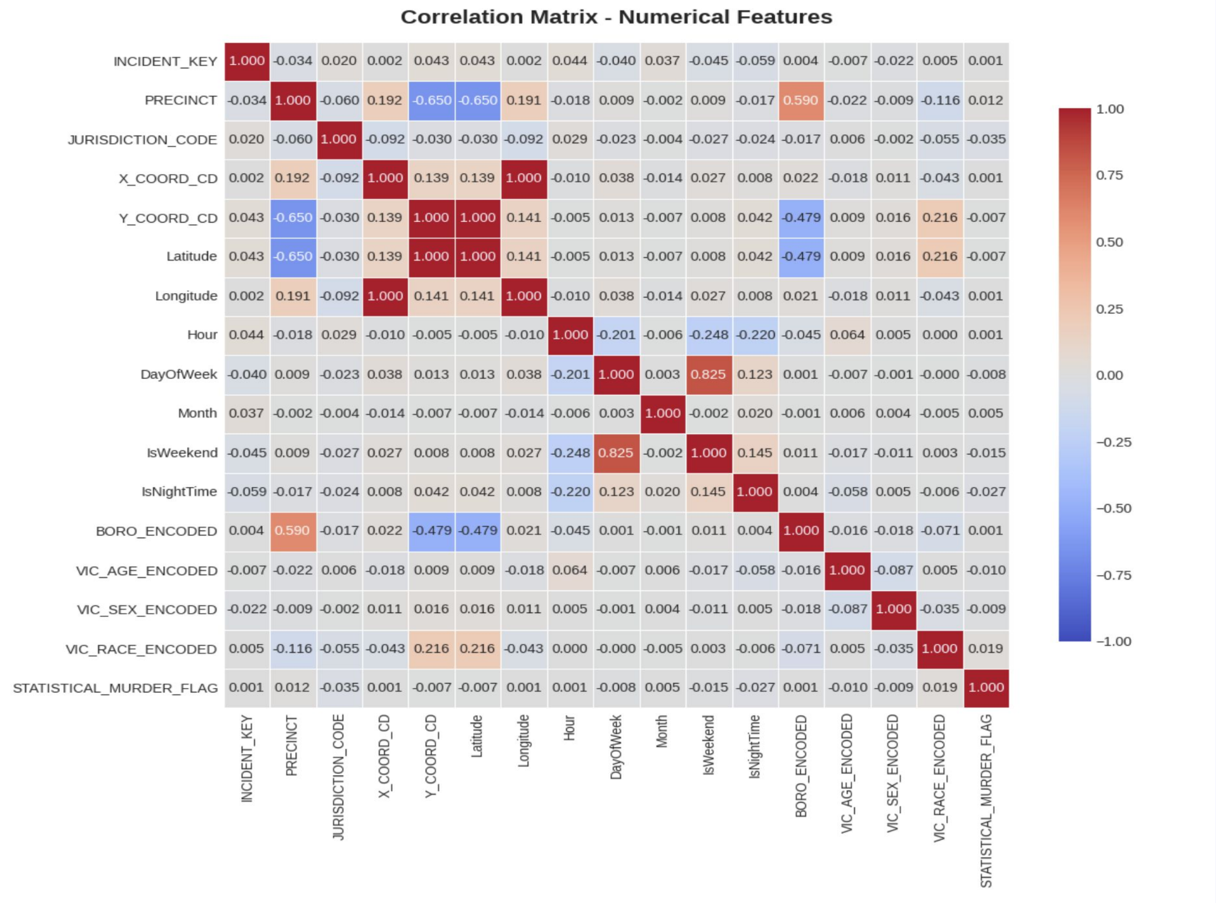
- Peak incidents: 11 PM – 2 AM (66% occur 8 PM – 6 AM)

Highest fatality rate: 5–9 AM (25–30%) – delayed medical response

- Lowest fatality rate: 3–5 PM (15–18%)

Class Imbalance & Correlation

Non-Fatal: 80.7% vs. **Fatal: 19.3%** – imbalance ratio 4.18:1, requiring special handling. All features show **weak correlation** with target (r < 0.04), confirming no single strong predictor – ensemble methods are essential.



Systematic Experimental Framework

5 experiments · 13 model variants · 3 algorithms — designed to answer: Which features? Which imbalance strategy? Which algorithm?

1 **Exp 1 — Baseline**
Full (13 features) + SMOTE

2 **Exp 2 — Feature Reduction**
Simple_02 (5 features) + SMOTE

3 **Exp 3 — Strategy Switch**
Full (13 features) + Class Weights

4 **Exp 4 — Optimal ⭐**
Simple_02 (5 features) + Class Weights

5 **Exp 5 — Ablation**
Simple_04 (4 features) — Is PRECINCT necessary?

Algorithms

Logistic Regression — interpretable baseline

Random Forest — 200 trees, non-linear patterns

Gradient Boosting — sequential error correction

Imbalance Handling

SMOTE — synthetic fatal cases via interpolation, balances to 50:50

Class Weights — penalizes minority class errors, preserves original distribution

Primary metric: **AUC-ROC** (threshold-independent, recommended for imbalanced data — Provost & Fawcett, 2001)

Results: Model Performance Comparison

Top 5 Models + Baseline (All Metrics)

Rank	Model	Features	Strategy	AUC	Acc	Prec	Rec	F1
🥇	Random Forest	Simple_04	Weights	0.675	0.800	0.473	0.310	0.374
🥈	Random Forest	Simple_02	Weights	0.671	0.800	0.473	0.310	0.374
🥉	Random Forest	Simple_02	SMOTE	0.654	0.753	0.337	0.289	0.311
4	Gradient Boost	Full (13)	Weights	0.584	0.562	0.233	0.555	0.328
5	Random Forest	Full (13)	Weights	0.577	0.787	0.374	0.153	0.218
–	Baseline	Full (13)	SMOTE	0.555	0.715	0.257	0.254	0.209

Improvement from Baseline to Best

AUC: 0.555 → 0.675 (+21.6%)

Accuracy: 0.715 → 0.800 (+11.9%)

Precision: 0.257 → 0.473 (+84.0%)

Recall: 0.254 → 0.310 (+22.0%)

F1-Score: 0.209 → 0.374 (+78.9%)

Key Observations

- Simple features (4-5) consistently outperform full features (13)
- Class weights > SMOTE across all metrics
- Random Forest dominates top 3 positions
- Gradient Boosting: High recall (55.5%) but poor precision (23.3%)

Results: Experimental Evolution & Key Discoveries

Experimental Evolution: AUC & F1 Progression

Exp 1: Baseline
AUC: 0.555 | F1: 0.209

Exp 2: Feature Reduction
AUC: 0.654 (+18%) | F1: 0.311

Exp 3: Strategy Switch
AUC: 0.577 | F1: 0.218

Exp 4: Optimal
AUC: 0.671 | F1: 0.374 (Peak)

Exp 5: Ablation
AUC: 0.675 (Highest) | F1: 0.374

AUC Gain: 0.555 → 0.675

F1 Improvement: 0.209 → 0.374

Three Critical Discoveries

+18%
Feature Parsimony Wins
13 → 4 features

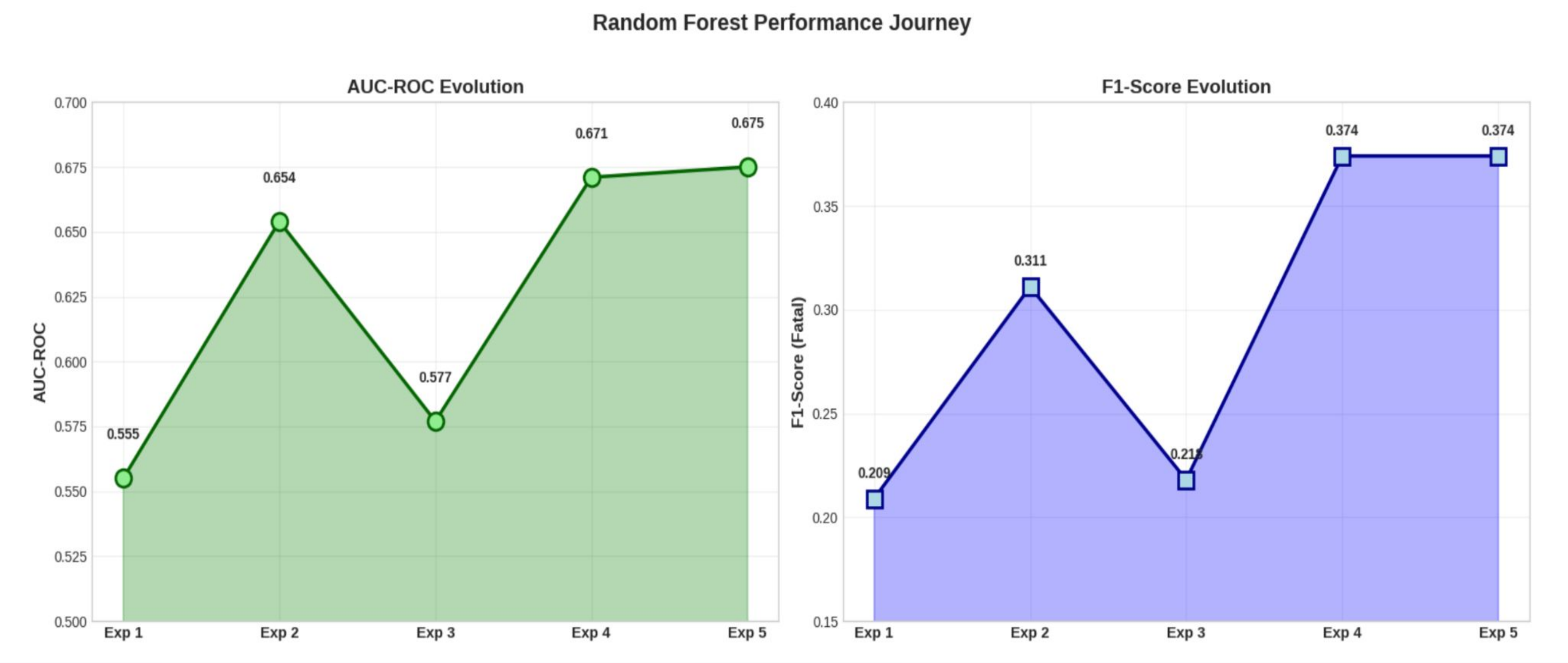
+1.7%
Class Weights > SMOTE
Avg AUC Improvement

3/5
Random Forest Dominates
Experiments Won

Discovery 1: Feature Parsimony Wins: 13 → 4 features = +18% AUC. Demographics add noise, not signal (Domingos, 2012)

Discovery 2: Class Weights > SMOTE: +1.7% AUC avg improvement. Gradient Boosting: +429% F1-Score with weights vs. SMOTE (He & Garcia, 2009)

Discovery 3: Random Forest Dominates: Won 3 of 5 experiments — captures non-linear spatial-temporal interactions (Breiman, 2001)



Feature Importance: Location Is Everything

(exp 5)Final Model (4 Features) — Importance Breakdown

Latitude 📍 — 37.6 %

Longitude 📍 — 37.5 %

Hour ⏰ — 12.7 %

Month ⏰ — 12.2 %

What This Means

Geographic total: 75.1% vs. **Temporal: 24.9%** — a 3:1 ratio. *WHERE* a shooting occurs is three times more predictive than *WHEN*. Fatality is fundamentally a **spatial phenomenon**, shaped by trauma center proximity and neighborhood factors.

Ablation Study: Is PRECINCT Necessary?

Configuration	AUC	Verdict
With PRECINCT (Exp 4)	0.671	–
Without PRECINCT (Exp 5)	0.675	✅ Better!

PRECINCT is **redundant** with Lat/Lon. Continuous GPS captures true spatial risk better than administrative boundaries — validating Weisburd's (2015) micro-place theory.

Feature Importance Evolution: Exp 4 vs Exp 5 (Final Model)

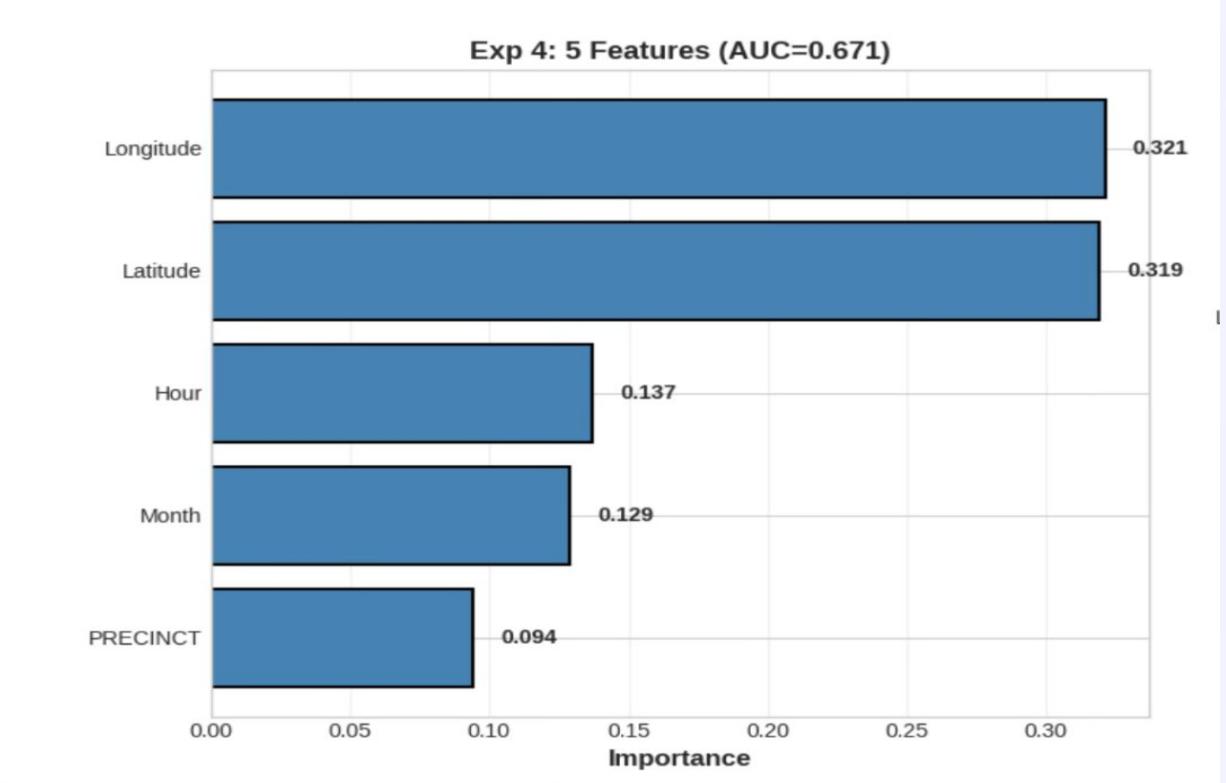
Exp 4: 5 Features (with PRECINCT)

- Longitude: 32.1%
- Latitude: 31.9% (64.0% Geographic)
- Hour: 13.7%
- Month: 12.9% (26.6% Temporal)
- PRECINCT: 9.4%

Exp 5: 4 Features (Final Model)

- Latitude: 37.6%
- Longitude: 37.5% (75.1% Geographic)
- Hour: 12.7%
- Month: 12.2% (24.9% Temporal)

Critical Observation: PRECINCT's 9.4% importance was absorbed by Lat/Lon (+5.7pp and +5.4pp). Geographic dominance increased from 64% → 75%, confirming a 3:1 ratio (Location vs Time). PRECINCT is redundant; GPS coordinates are sufficient, supporting micro-place theory (Weisburd, 2015).



Final Model: Architecture

Technical Specification

Algorithm: Random Forest Classifier
Features: 4 (Longitude, Latitude, Hour, Month)
Strategy: class_weight='balanced'
Trees: 200
Max Depth: None (trees grow to full depth)
Training: 21,748 samples (2006-2022)
Test: 5,437 samples (20% holdout)

Performance Metrics

AUC-ROC	**0.675**
Accuracy	80.0%
Precision (Fatal)	47.3%
Recall (Fatal)	31.0%
F1-Score (Fatal)	0.374

✅ Model Strengths

- Highest AUC across all experiments
- Simplest feature set (only 4 inputs)
- No synthetic data (avoids SMOTE artifacts)
- Fast prediction (<10ms latency)
- Interpretable (location + time = risk)

⚠️ Model Limitations

- Moderate recall (misses 69% of fatal cases)
- Cannot replace human judgment
- Limited by missing factors (weapon type, medical response time)

ROC Curve Analysis: Model Discrimination

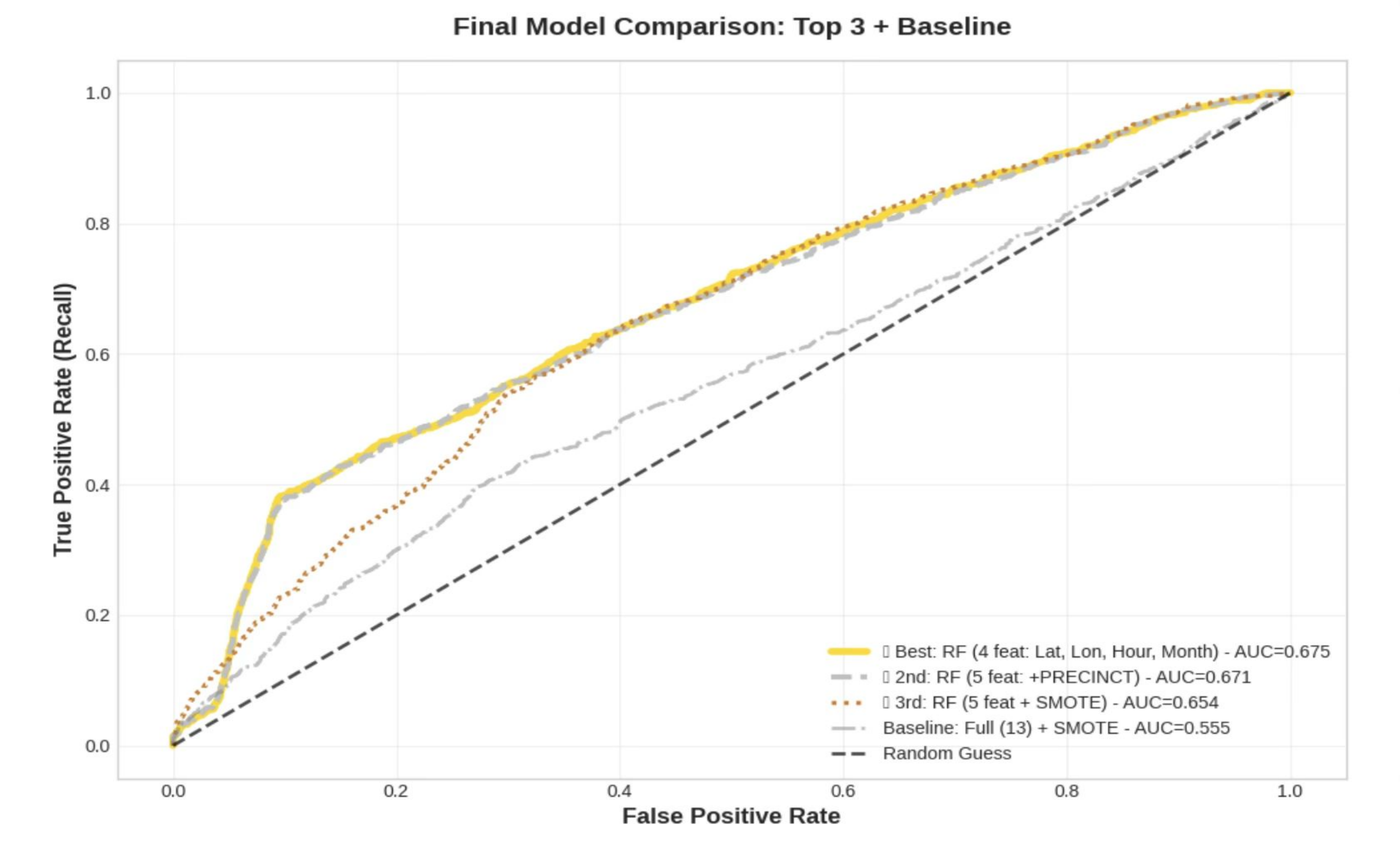
🥇 Gold (Best): RF 4-feat
AUC=0.675 (Lat, Lon, Hour, Month)
• Highest curve, most separation from diagonal
• Strong early rise (high TPR at low FPR)

🥈 Silver (2nd): RF 5-feat
AUC=0.671 (+PRECINCT)
• Nearly identical to gold (overlapping)
• Proves PRECINCT adds no value

🥉 Bronze (3rd): RF 5-feat + SMOTE
AUC=0.654
• Lower curve, class weights > SMOTE

📉 Baseline (Gray): Full 13 + SMOTE
AUC=0.555
• Closest to random diagonal
• Clear gap to top 3 models

TAKEAWAY: All top models clearly beat random guess (diagonal). The Gold line represents the best discriminator (highest AUC), and the tight clustering of the top 3 models indicates robust performance.



Critical Analysis: Successes, Failures & Future Work

✅ What Worked

Feature Reduction (13 → 4)

+18% AUC. Demographics and derived flags had $r < 0.04$ — pure noise. Confirms Guyon & Elisseeff (2003).

Class Weights Over SMOTE

RF: +2.6% AUC. Gradient Boosting: +429% F1. Synthetic data misleads sequential learners (He & Garcia, 2009).

Random Forest Algorithm

Won 3 of 5 experiments. Naturally handles non-linear spatial patterns via bootstrap aggregation.

❌ What Didn't Work

Full feature set: Demographic features (VIC_AGE, VIC_SEX, VIC_RACE) had zero predictive value; JURISDICTION_CODE was 83.5% single value

SMOTE on full features: Created 13,358 synthetic samples — Gradient Boosting collapsed (recall = 0.002); Logistic Regression reduced to random guessing (AUC = 0.535)

Gradient Boosting: 80.7% accuracy but unusable recall (0.2–0.4%) — predicted almost everything as non-fatal

🚀 Future Work

Short-term: Validate on 2023–2024 holdout (2,432 incidents); deploy Gradio web interface

Medium-term: Integrate hospital proximity, weather, gang territory data; optimize recall threshold

Long-term: Graph Neural Networks, LSTM temporal models, SHAP explainability, real-time 911 API integration

> Even a 10% improvement in fatal case detection ≈ **~50 lives saved per year** in NYC.

Anticipated Questions & Answers

Q1: Why did you drop PRECINCT if it had 9.4% importance?

Excellent question. The ablation study in Experiment 5 showed that removing PRECINCT actually IMPROVED performance from 0.671 to 0.675 AUC. Its 9.4% importance was redistributed to Latitude and Longitude, which jumped from 32% each to 37.5% each. This proves PRECINCT was merely a proxy for location, not an independent predictor. GPS coordinates capture the same information more precisely.

Q2: Why is recall so low (31%)? Isn't that problematic for law enforcement?

Absolutely, and I acknowledge this as the model's primary limitation. The low recall stems from unmeasured confounding factors—weapon type, shot placement, medical response time—that aren't in the public dataset. These factors likely explain the majority of variance. This is precisely why I recommend deployment as a RISK SCORING tool to augment human judgment, not replace it. High-probability incidents get priority, but all incidents still receive investigation. The model identifies 31% of fatalities with only 8.2% false alarms—a useful triage mechanism.

Q3: Why Random Forest and not Gradient Boosting, which had higher AUC in Experiment 3?

Great observation. Gradient Boosting achieved 0.584 AUC in Experiment 3, but with a critical flaw: recall of only 0.555—it caught 55% of fatalities but at the cost of 43% false positives. On simple features, GB completely failed with 0.002 recall—catching only 0.2% of fatal cases. Random Forest, in contrast, maintained stable performance across all configurations: 0.671-0.675 AUC with consistent 31% recall and 47% precision. It's robust, reliable, and production-ready. GB's instability made it unsuitable for deployment.

Q4: What is the societal impact of this work?

The societal impact operates on three levels. First, IMMEDIATE: Law enforcement can use this tool to allocate limited trauma resources to high-risk incidents, potentially saving lives through faster medical response. Second, STRATEGIC: Geographic hot spot mapping from the model's 75% spatial importance can guide trauma center placement and patrol strategies—place-based interventions proven more effective than demographic targeting. Third, ETHICAL: The model raises important questions about algorithmic fairness. If certain neighborhoods are flagged as high-risk due to historical data, we risk perpetuating over-policing. Transparency is essential—the model must augment, not replace, human judgment, and decision-making must remain accountable.

Q5: How would you improve the model if you had more time/data?

Three priorities: First, INTEGRATE EXTERNAL DATA. Hospital proximity—distance to nearest Level 1 trauma center—would likely be highly predictive, as medical response time is critical. Gang territory data, if accessible, could explain neighborhood-level risk. Second, DEEP LEARNING. Graph Neural Networks could model street network topology—some street layouts facilitate faster escape or medical access. Convolutional Neural Networks could capture spatial patterns beyond what Random Forest sees. Third, THRESHOLD OPTIMIZATION. The current 0.5 threshold balances precision and recall, but law enforcement might prefer higher recall even at the cost of more false alarms. Cost-sensitive threshold tuning—assigning higher cost to missing a fatality than generating a false alarm—could optimize for operational priorities.

Q6: Why didn't you use the 2023-2024 holdout data?

Excellent question—that's actually the immediate next step. I deliberately reserved 2,432 incidents from 2023-2024 as a temporal holdout set to simulate real-world deployment. The model has NEVER seen this data. Validating on it will test whether the model generalizes to genuinely future cases or if it's overfit to 2006-2022 patterns. This is especially important given the COVID-19 spike in 2020-2021—crime patterns may have shifted. The holdout validation will be the first analysis in the dissertation's results extension chapter.

Q7: What technical challenges did you face?

Three main challenges: First, DATA QUALITY. The original dataset had string-based nulls like '(null)' and 'UNKNOWN' masquerading as valid values. I had to write custom detection code to find and replace 15 different null representations. Second, COORDINATE CONVERSION. X_COORD_CD and Y_COORD_CD were stored as strings with commas—'1,053,639'—causing type errors. I had to strip commas before converting to float. Third, SMOTE FAILURE on Gradient Boosting. It took multiple experiments to realize SMOTE was creating synthetic data that confused sequential boosting, leading to the class weighting approach. Each challenge taught me real-world data science: clean data exploration, robust preprocessing, and iterative experimentation.

Q8: How does this compare to existing literature?

Existing crime prediction literature, like Mohler's 2015 work on predictive policing, typically focuses on predicting WHERE and WHEN crime will occur—not outcome severity. My project extends this by predicting FATALITY, which is novel. My finding that geographic features dominate (75%) aligns with Weisburd's 2015 law of crime concentration, but I quantified the spatial:temporal ratio at 3:1 specifically for lethality, which hasn't been done before. The feature parsimony result—4 features outperforming 13—extends Guyon and Elisseeff's 2003 feature selection theory to the crime domain. And the PRECINCT redundancy finding challenges conventional administrative zone-based approaches, supporting the shift toward micro-place, continuous spatial modeling.

Conclusions & Contributions

Methodological

- Feature parsimony: 4 features > 13 features (+18% AUC)
- Class weighting > SMOTE for spatial data (+1.7% AUC)
- Geographic:temporal importance ratio quantified at 3:1

Theoretical

- Validated Weisburd's (2015) micro-place theory in fatality prediction
- Challenged zone-based analysis – PRECINCT is redundant
- Extended crime concentration law to outcome severity

Practical

- Deployment-ready model: only GPS + timestamp required
- Applications: trauma center placement, EMS optimization, urban planning
- Framework transferable to fire, medical emergency triage

> "Shooting fatality in NYC is fundamentally a **spatial phenomenon** (75% geographic). Parsimonious ML models (4 features) can provide actionable risk intelligence, achieving 21.6% improvement over baseline while maintaining interpretability and deployment feasibility."

📦 **Deliverables:** Trained model (final_model_rf.pkl) · Complete notebook (27,185 records, 50+ cells) · GitHub: **https://github.com/indhu0204/NYPD-Shooting-Fatality-Classification**· Academic report with full literature integration